

Testing Approaches

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graph TD; A[Testing Approaches] --> B[Static]; A --> C[Dynamic]; A --> D[Fault Injection]; B --> B1[Formal Verification]; B --> B2[Static Timing Analysis]; B --> B3[Design Rule Check (DRC)]; B --> B4[Lint / CDC Check]; C --> C1[Functional Simulation]; C --> C2[Built-In Self-Test (BIST)]; C --> C3[Boundary Scan (JTAG)]; C --> C4[Hardware-in-the-Loop]; D --> D1[SEU Emulation]; D --> D2[Radiation Testing]; D --> D3[Laser / Pin Injection]; D --> D4[SW-Based Fault Injection];
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Static

Formal Verification

Static Timing Analysis

Design Rule Check (DRC)

Lint / CDC Check

Dynamic

Functional Simulation

Built-In Self-Test (BIST)

Boundary Scan (JTAG)

Hardware-in-the-Loop

Fault Injection

SEU Emulation

Radiation Testing

Laser / Pin Injection

SW-Based Fault Injection